

Artificial Intelligence in Modern Business

Artificial intelligence is increasingly being used by companies to automate repetitive tasks, improve customer service, analyze large amounts of data, and support business decisions.

One major use of AI is customer support. Companies use AI assistants and chatbots to answer common customer questions, summarize support tickets, route requests to the correct team, and provide faster responses. These systems can reduce the workload on human agents while allowing people to focus on more complex issues.

Another important application is fraud detection. Banks and financial institutions use machine learning models to analyze transaction patterns. These models can identify unusual behavior, such as an unexpected purchase location, a sudden increase in transaction amount, or many transactions occurring within a short period of time. When suspicious activity is detected, the transaction may be flagged for further review.

AI is also used in healthcare. Machine learning systems can assist doctors by analyzing medical images, patient records, and laboratory results. These systems are designed to support medical professionals rather than replace them. Human review is especially important because healthcare decisions can have serious consequences.

Retail companies use recommendation systems to suggest products to customers. These systems analyze past purchases, browsing behavior, and similarities between users or products. For example, an online store may recommend a laptop bag to someone who recently purchased a laptop.

Large language models can also be connected to company documents using Retrieval-Augmented Generation, commonly called RAG. In a RAG system, documents are divided into smaller chunks and converted into embeddings. The embeddings are stored in a vector database. When a user asks a question, the system converts the question into an embedding and retrieves the most relevant document chunks. Those chunks are then provided to a language model as context so that it can generate a grounded answer.

RAG is useful because organizations can provide domain-specific information to a language model without retraining the entire model. It can also make it easier to update the knowledge available to the application because documents can be added or changed independently of the underlying language model.

However, AI systems still require careful evaluation, monitoring, security, and human oversight. Models can make mistakes, retrieve irrelevant information, or generate inaccurate answers. Production AI systems therefore need mechanisms for testing quality, monitoring failures, protecting sensitive information, and controlling access to tools and data.